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Corso di Laurea in Informatica

Appunti Advanced Machine Learning

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Chapter 1

Introduction to Deep Learning

Deep learning is a specialized subset of machine learning centered on multilayer neural networks, whose architectures are conceptually inspired by the structure of the human brain. Rather than relying on hand-crafted features, deep learning operates within the paradigm of *representation learning*, enabling models to automatically discover hierarchies of abstract features built from simpler ones. Today, deep learning underpins many of the most advanced artificial intelligence (AI) systems, driving state-of-the-art applications ranging from computer vision and generative AI to autonomous driving and robotics. As illustrated in Figure 1.1, deep learning represents a specific, highly expressive branch within the broader scope of artificial intelligence.

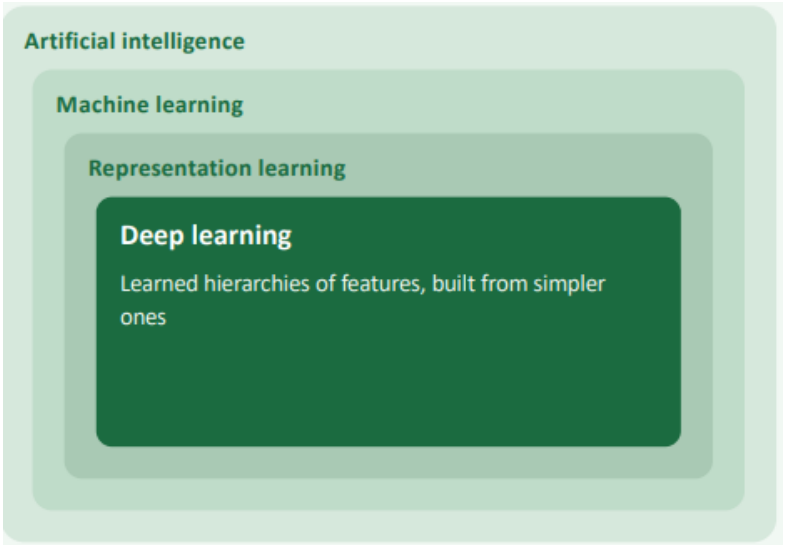


Figure 1.1: Hierarchical positioning of Deep Learning

1.0.1 Representation learning

What really distinguishes deep learning from traditional machine learning is its ability to perform *representation learning*. In traditional machine learning, the model is provided with a set of features that are manually engineered by domain experts. In contrast, deep learning models

automatically learn to extract relevant features from raw data with representation learning. In figure 1.2 we can see four different approaches to the task of representation. The main goal of

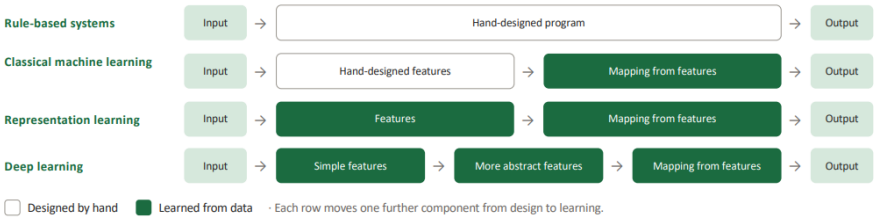


Figure 1.2: Four approaches to representation learning

representation learning is to identify the **factor of variations** that explain the observed data (that sometimes are not directly observable). They might be seen as concepts or abstraction that help us make sense of the rich variability in the data.

1.0.2 Deep learning

Deep learning uses a hierarchical approach to representation learning, allowing the models obtain features by introducing representations that are expressed in therms of other simpler representations.